

■ 1.5.4 Sums and Products

This constructs the sum $\sum_{i=1}^7 \frac{x^i}{i}$.

```
In[1]:= Sum[x^i/i, {i, 1, 7}]
Out[1]= x +  $\frac{x^2}{2}$  +  $\frac{x^3}{3}$  +  $\frac{x^4}{4}$  +  $\frac{x^5}{5}$  +  $\frac{x^6}{6}$  +  $\frac{x^7}{7}$ 
```

You can leave out the lower limit if it is equal to 1.

```
In[2]:= Sum[x^i/i, {i, 7}]
Out[2]= x +  $\frac{x^2}{2}$  +  $\frac{x^3}{3}$  +  $\frac{x^4}{4}$  +  $\frac{x^5}{5}$  +  $\frac{x^6}{6}$  +  $\frac{x^7}{7}$ 
```

This makes i increase in steps of 2, so that only odd-numbered values are included.

```
In[3]:= Sum[x^i/i, {i, 1, 5, 2}]
Out[3]= x +  $\frac{x^3}{3}$  +  $\frac{x^5}{5}$ 
```

Products work just like sums.

```
In[4]:= Product[x + i, {i, 1, 4}]
Out[4]= (1 + x) (2 + x) (3 + x) (4 + x)
```

<code>Sum[f, {i, imin, imax}]</code>	the sum $\sum_{i=imin}^{imax} f$
<code>Sum[f, {i, imin, imax, di}]</code>	the sum with i increasing in steps of di
<code>Sum[f, {i, imin, imax}, {j, jmin, jmax}]</code>	the nested sum $\sum_{i=imin}^{imax} \sum_{j=jmin}^{jmax} f$
<code>Product[f, {i, imin, imax}]</code>	the product $\prod_{i=imin}^{imax} f$

Sums and products.

Mathematica gives an exact result for this sum.

```
In[5]:= Sum[1/i^3, {i, 1, 20}]
Out[5]=  $\frac{336658814638864376538323}{280346265322438720204800}$ 
```

Here is the numerical value.

```
In[6]:= N[%]
Out[6]= 1.20087
```

Mathematica cannot give you an exact result for this infinite sum.

```
In[7]:= Sum[1/i^3, {i, 1, Infinity}]
Out[7]= Sum[i-3, {i, 1, Infinity}]
```

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You can still get a numerical result.

```
In[8]:= N[ % ]
Out[8]= 1.20206
```

Mathematica also has a notation for multiple sums and products.

`Sum[f, {i, imin, imax}, {j, jmin, jmax}]` represents a sum over i and j , which would be written in standard mathematical notation as $\sum_{i=imin}^{imax} \sum_{j=jmin}^{jmax} f$. Notice that in *Mathematica* notation, as in standard mathematical notation, the range of the *outermost* variable is given *first*.

This is the multiple sum $\sum_{i=1}^3 \sum_{j=1}^i x^i y^j$.

Notice that the outermost sum over i is given first, just as in the mathematical notation.

```
In[9]:= Sum[x^i y^j, {i, 1, 3}, {j, 1, i}]
Out[9]= x y + x^2 y + x^3 y + x^2 y^2 + x^3 y^2 + x^3 y^3
```

The way you specify the ranges of variables in `Sum` and `Product` is an example of the rather general *iterator notation* that *Mathematica* uses. You will see this notation again when we discuss generating tables and lists using `Table` (Section 1.7.2), and when we describe `Do` loops (Section 2.5.2). The function `Range` also uses essentially the same notation.

<code>{imax}</code>	iterate $imax$ times, without incrementing any variables
<code>{i, imax}</code>	i goes from 1 to $imax$ in steps of 1
<code>{i, imin, imax}</code>	i goes from $imin$ to $imax$ in steps of 1
<code>{i, imin, imax, di}</code>	i goes from $imin$ to $imax$ in steps of di
<code>{i, imin, imax}, {j, jmin, jmax}</code>	i goes from $imin$ to $imax$, and for each value of i , j goes from $jmin$ to $jmax$

Mathematica iterator notation.